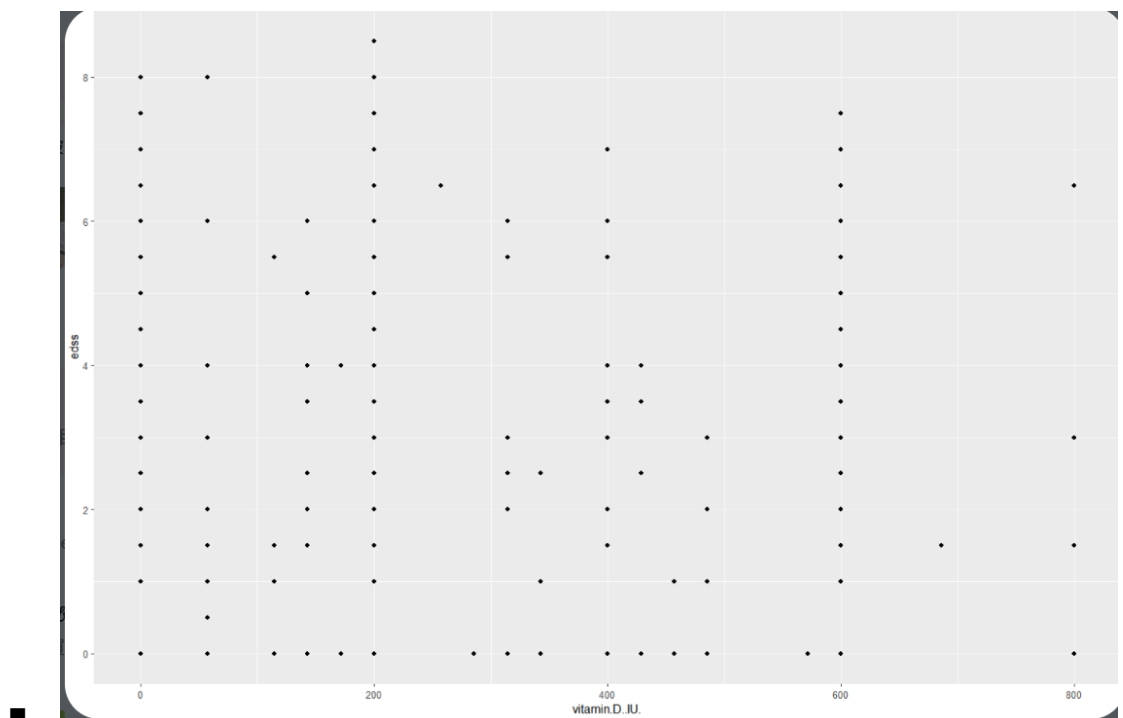


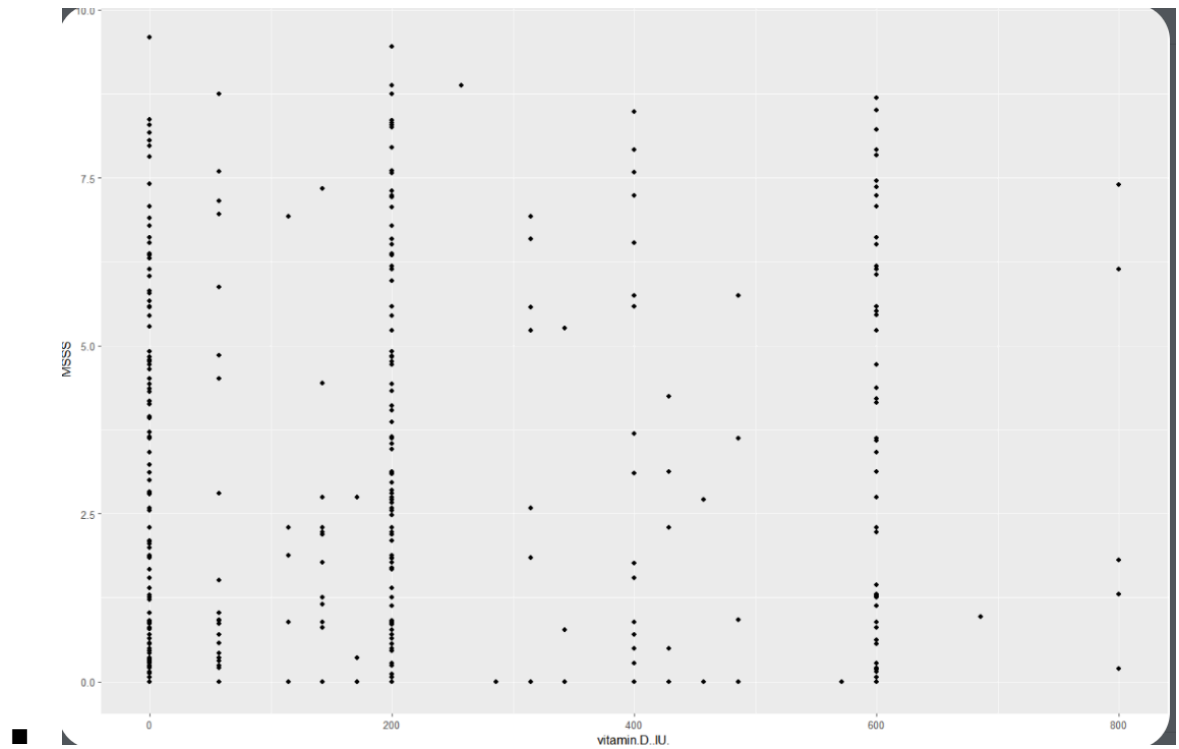
Meeting note:

Group assignment reflection:

- Research aim: not really worded as question
 - Available clear action verbs to use: assess, investigate, correlate, analyze, etc
- Vitamin D, microbiome, and MS severity appear to be discussed as separated components currently, need to clarify the relationship in between feature
 - Vitamin D → gut microbiome → MS severity
- Taxonomy bar plot
 - Useful for background overview, not for addressing the main aims
 - Provide visualization of
 - phylum/family level composition
 - General increase/decrease trend in MS patient
 - Can serve as sanity check for expected MS-associated taxa patterns
- Straifying vitamin D
 - Categorize in the background analysis
 - Could group into 0 IU, >0 & ≤200 IU, >200 IU (depends on visualization)
 - Equal sample size per group is ideal, but not necessary
- MSSS/EDSS vs Vitamin D (supplementary analysis)
 - Dot plot:
 - Vitamin D intake vs EDSS

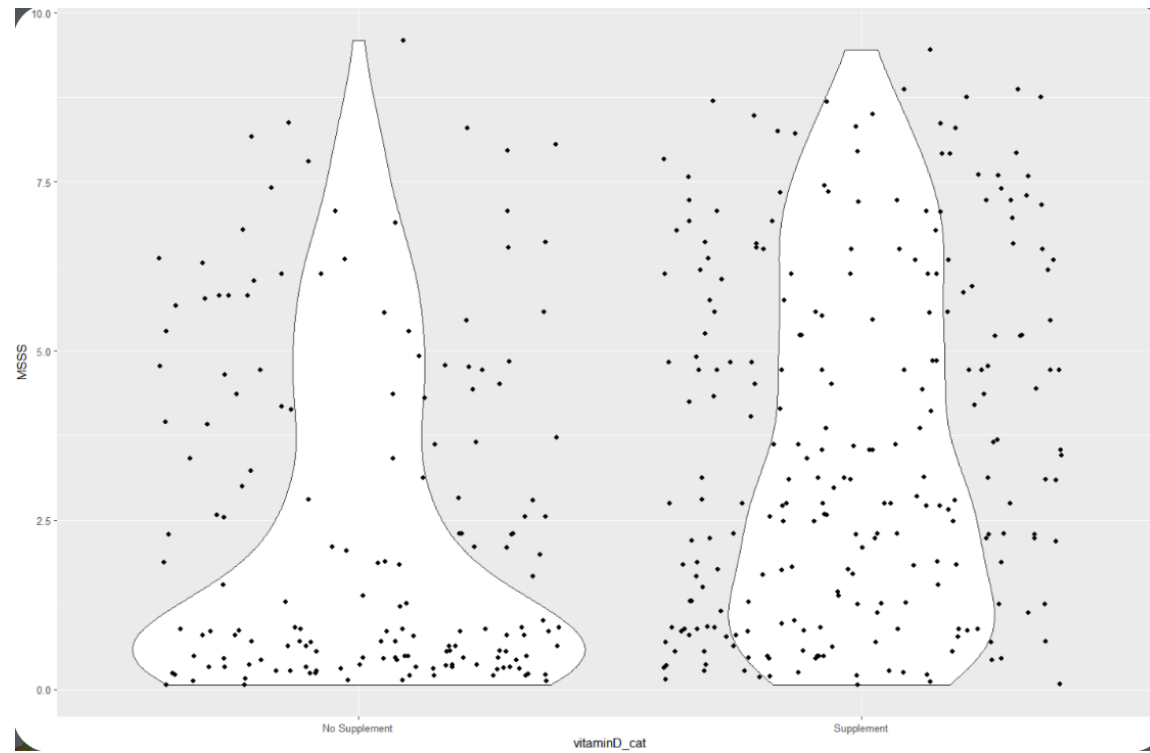


■ Vitamin D intake vs MSSS

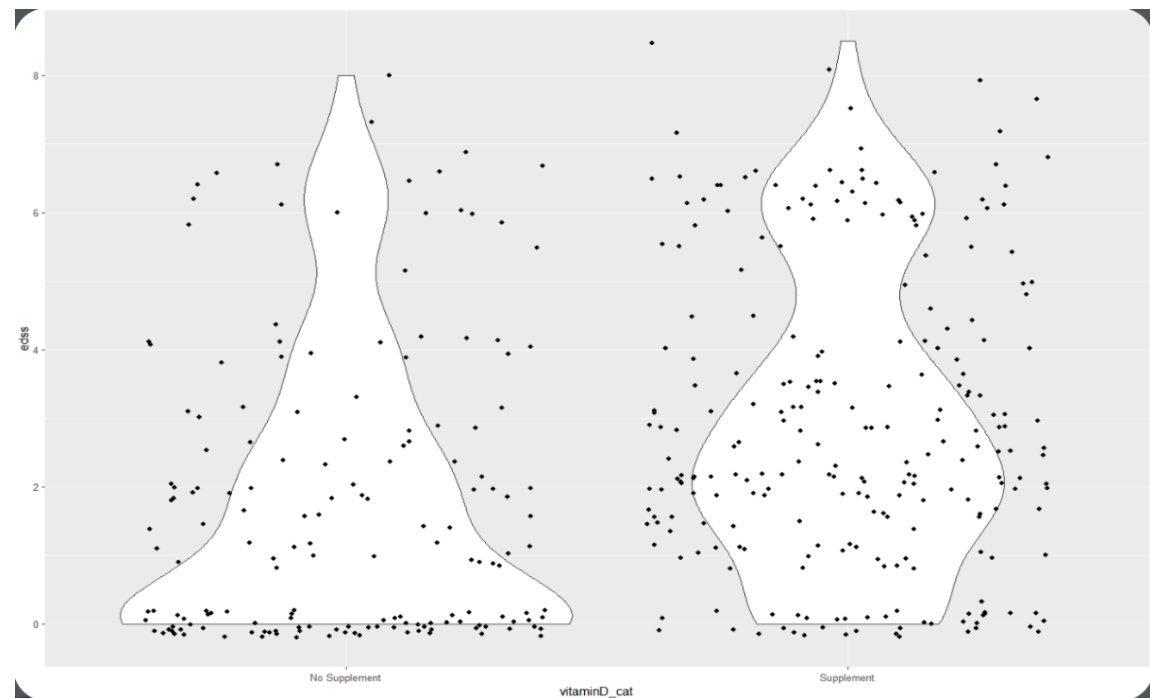


- Determine which severity score correlates better with vitamin D
- The analysis can be included as supplementary figures
- Violin plot:
 - Remove non-MS patients
 - Categorized vitamin D into 0 and non-0
 - Layout individual data points (recommend)

■ Vitamin D intake vs MSSS



■ Vitamin D intake vs EDSS



- Note: usual vitamin D intake amount recommendation: 600-800 IU
 - Supplement intake (our data) may not reflect on biological level
 - Interpretation could acknowledge this limitation

Feedback on aims:

- Second aim: separate into different aim
 - Random forest, better to fit into the 4th aim
 - Predict vitamin D: good sanity check, if can repredict the vitamin D level, indicate the model has better accuracy
 - E.g. Vitamin D associated microbiome correlate with ms patient
- Third aim:
 - Use PICRUSt2 to predict metabolic pathways based on 16S rRNA data
 - Be aware that 16S data can only predict functional potential, not confirm actual metabolic activity
 - Clearly state “predicted functional pathway” in the analysis to reflect this limitation
- Fourth aim: can be clarified further
 - Identify candidate markers using:
 - Random forest
 - Differential abundance analysis
 - Indicator species analysis
 - Evaluate each marker individually by fitting it to MS severity (e.g. MSSS/EDSS) and assess predictive performance
 - Build a combined model using top markers and compare its performance to individual-marker models

Proposal:

- Start drafting background early, build structure skeleton
- Rationalize the reasoning of studying this research question
 - Include evidence from literature suggesting correlation
- Use primary paper for literature review
 - Paper: not limited to the dataset used in our research

To-do:

- Reword research aims clearly
- Separate aim 2 and aim 4
- Conduct alpha and beta diversity analysis
- Draft proposal